

SFWRENG-3MX3 Midterm 2 Practice

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1) Short Questions

1.a) What are the coefficients X_k of the Fourier series of the signal $x(t) = \cos(t) + \sin(2t)$.

Solution:

Looking at the frequency content of each term of $x(t)$, and noting that $\cos(t) = \frac{1}{2}(e^{it} + e^{-it})$ and $\sin(t) = \frac{1}{2i}(e^{it} - e^{-it})$, we can see that:

$$X_{-2} = \frac{-1}{2i}$$

$$X_{-1} = \frac{1}{2}$$

$$X_0 = 0$$

$$X_1 = \frac{1}{2}$$

$$X_2 = \frac{1}{2i}$$

1.b) For a real-valued system, we know $H\left(\frac{\pi}{4}\right) = 1 + i$. What is the value of $H\left(\frac{-\pi}{4}\right)$ and $H\left(\frac{9\pi}{4}\right)$?

Solution:

(I) What is $H\left(\frac{-\pi}{4}\right)$?

We know that the frequency response for all systems must have a symmetric conjugate complex, meaning that $H(-\omega) = \bar{H}(\omega)$. Therefore, since the complex conjugate of $1 + i$ is $1 - i$, $H\left(\frac{-\pi}{4}\right) = 1 - i$.

(II) What is $H\left(\frac{9\pi}{4}\right)$?

*Note here that the argument I'm about to supply works only for the impulse response of discrete systems, since it's not specified, I'm not entirely sure why this solution is *always* correct, or if it's merely an accidental omission from the problem statement.*

Notice here that $H\left(\frac{9\pi}{4}\right) = H\left(2\pi + \frac{\pi}{4}\right)$. Thus, since the frequency response of a discrete system is always 2π -periodic, we know that $H(\omega) = H(\omega + 2\pi)$ and thus $H\left(2\pi + \frac{\pi}{4}\right) = H\left(\frac{\pi}{4}\right) = 1 + i$.

2) The Frequency Response

If the frequency response of a system is given by $H(\omega) = \cos(\omega)$, compute the output of the system for the input signal $x(n) = 2 + \cos(\frac{\pi}{2}n + \frac{\pi}{4}) + \sin(\pi n + \frac{3\pi}{2})$.

Solution:

Noting that the frequency of each term of the input signal is 0 , $\frac{\pi}{2}$, and π , respectively, we can find that:

ω	$H(\omega)$	$ H(\omega) $	$\arg(H(\omega))$
0	1	1	0
$\frac{\pi}{2}$	0	0	$-$
π	-1	1	0

Therefore, we know that:

Verify if this is correct

$$\begin{aligned}
 y(n) &= |H(\omega_1)| \cos(\omega_1 n + \arg(H(\omega_1))) + |H(\omega_2)| \cos(\omega_2 n + \arg(H(\omega_2))) + |H(\omega_3)| \cos(\omega_3 n + \arg(H(\omega_3))) \\
 &= 1 \cos(0 + 0) + 1 \cos(\pi n + 0) \\
 &= 1 + \cos(\pi n)
 \end{aligned}$$

3) Filter Design

Give the difference equation of a causal, minimum order, discrete, real-valued system that removes any signal with the normalized frequency of 0 and $\frac{\pi}{2}$ and passes signals a signal with a normalized frequency of π with a gain of 1.

Solution:

To solve this problem, we must first construct a prototype filter $\tilde{H}(\omega)$ which satisfies $\tilde{H}(0) = 0$ and $\tilde{H}(\frac{\pi}{2}) = 0$ and then normalize it to produce our final filter $H(\omega)$. Using this final frequency response corresponding to our filter, we can then recover the difference equation of the system.

(I) Find $\tilde{H}(\omega)$.

$$\begin{aligned}\tilde{H}(\omega) &= (1 - e^{-i\omega})(i + e^{-i\omega})(i - e^{-i\omega}) \\ &= (1 - e^{-i\omega})(-1 - e^{-2i\omega}) \\ &= -1 - e^{-2i\omega} + e^{-i\omega} + e^{-3i\omega}\end{aligned}$$

Checking, we see that:

$$\begin{aligned}|\tilde{H}(\pi)| &= |-1 - e^{-2i\pi} + e^{-i\pi} + e^{-3i\pi}| \\ &= |-1 - 1 - 1 - 1| \\ &= |-4| \\ &= 4\end{aligned}$$

(II) Find $H(\omega)$.

Since our prototype filter, $\tilde{H}(\omega)$, already has a gain of zero at 0 and $\frac{\pi}{2}$, we need only normalize it for $\omega = \pi$ to produce our final filter, which we can do by scaling $\tilde{H}(\omega)$ by a factor of $\frac{1}{4}$. Thus:

$$H(\omega) = \frac{1}{4}(-1 - e^{-i\omega} + e^{-2i\omega} + e^{-3i\omega})$$

(III) Find the difference equation for the filter.

By multiplying both sides by $e^{i\omega n}$, and recognizing that $y(n) = H(\omega)e^{i\omega n}$ when $x(n) = e^{i\omega n}$, we find the difference equation to be:

$$\begin{aligned}H(\omega)e^{i\omega n} &= \frac{1}{4}(-e^{i\omega n} - e^{-i\omega}e^{i\omega n} + e^{-2i\omega}e^{i\omega n} + e^{-3i\omega}e^{i\omega n}) \\y(n) &= \frac{1}{4}(-e^{i\omega n} - e^{i\omega(n-1)} + e^{i\omega(n-2)} + e^{i\omega(n-3)}) \\y(n) &= \frac{1}{4}(-x(n) - x(n-1) + x(n-2) + x(n-3))\end{aligned}$$

4) The Frequency Response

The system $y(n) + y(n-1) = 2x(n) - 5x(n-2)$ has a frequency response of $H(\omega) = \frac{1+e^{-i\omega}}{e^{-i\omega}}$. Compute the output of the system for the input $x(n) = \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + \sin\left(\pi n + \frac{3\pi}{2}\right)$.

Solution:

Noting that the frequency of each term of the input signal is $\frac{\pi}{2}$ and π respectively, we can find that:

ω	$H(\omega)$	$ H(\omega) $	$\arg(H(\omega))$
$\frac{\pi}{2}$	$\frac{1-i}{-i} = 1+i$	$\sqrt{2}$	$\frac{\pi}{4}$
π	0	0	-

Therefore, we know that:

$$\begin{aligned}
 y(n) &= |H(\omega_1)| \cos(\omega_1 n + \arg(H(\omega_1))) + |H(\omega_2)| \cos(\omega_2 n) \\
 &= \sqrt{2} \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right)
 \end{aligned}$$

5) The Fourier Transform

A Continuous Time Fourier Transform (CTFT) table states that:

$$x(t) = \begin{cases} \frac{\pi}{a}, & \text{if } -a \leq t \leq a \\ 0, & \text{otherwise} \end{cases} \iff X(\omega) = 2\pi \frac{\sin(a\omega)}{a\omega}$$

Use the table to compute the Fourier transform of the signal $x(t) = \begin{cases} 1, & \text{if } 0 \leq t \leq 2 \\ 0, & \text{otherwise} \end{cases}$.

Solution:

Consider the case when $a = 1$, then by the CTFT table, we know that:

$$x(t) = \begin{cases} \frac{\pi}{1}, & \text{if } -1 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases} \iff X(\omega) = 2\pi \frac{\sin(\omega)}{\omega}$$

Such a signal is nearly identical to the one we are requested to find in the question, except that it is scaled by a factor of π and our signal starts at a delay of 1 second afterwards (since it is only equal to 1 on the interval $0 \leq t \leq 2$ as opposed to $-1 \leq t \leq 1$). Therefore, if we scale this signal down by a factor of $\frac{1}{\pi}$ and delay it by a unit of time equal to $t = 1$, we can recover the Fourier transform of our given $x(t)$. Doing such a scaling and delaying is reflected in the Fourier transform as multiplying by $\frac{1}{\pi}$ and $e^{-i\omega}$ respectively. Therefore, the Fourier transform for our signal is given by:

$$\begin{aligned} X(\omega) &= 2\pi \frac{\sin(\omega)}{\omega} \cdot \frac{1}{\pi} \cdot e^{-i\omega n} \\ &= \frac{2e^{-i\omega n} \sin(\omega)}{\omega} \end{aligned}$$